

ALIE Byzantine Attack Comparison: CNN vs. SNN

Different Delta Values (Simplest Sweeps)

ByzFL SNN Benchmark

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1 Introduction

This document presents comparison grids (2x2 subfigures) comparing the Convolutional Neural Network (CNN) and Spiking Neural Network (SNN) models under the Optimal A Little Is Enough (ALIE) Byzantine attack. We compare four different step size (δ) parameters:

- **ALIE** $\delta = 10$ (Default positive large step size)
- **ALIE** $\delta = 1$ (Positive small step size)
- **ALIE** $\delta = -1$ (Negative small step size)
- **ALIE** $\delta = -10$ (Negative large step size)

Each comparison illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 5]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$) with a total of $N = 10$ honest clients.

2 ALIE Attack: Best Aggregator Choice

This section compares performance under ALIE using the dynamically selected best overall aggregator.

2.1 Convolutional Neural Network (CNN)

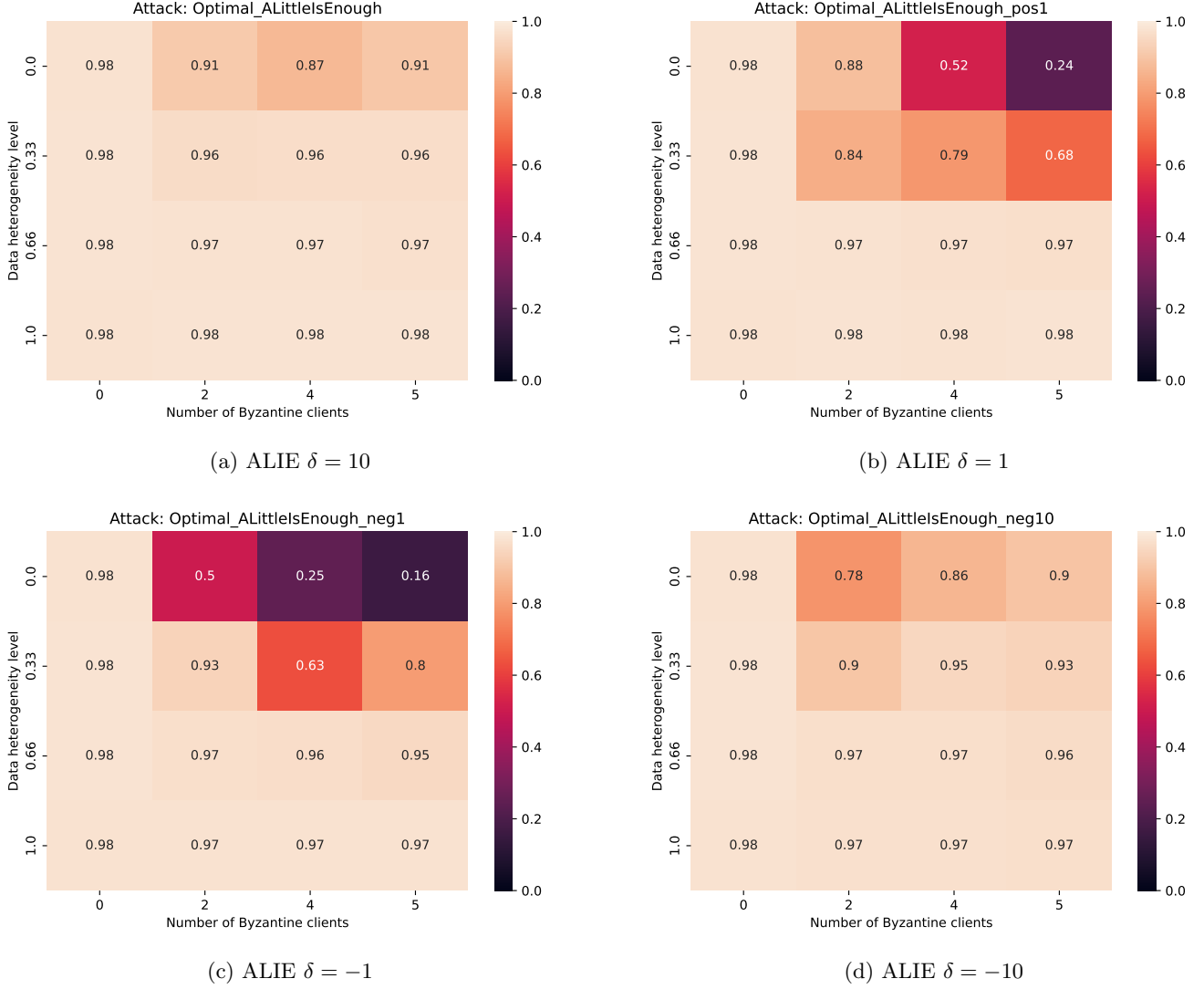


Figure 1: CNN accuracy comparison under ALIE attacks using the best overall aggregator selection.

2.2 Spiking Neural Network (SNN)

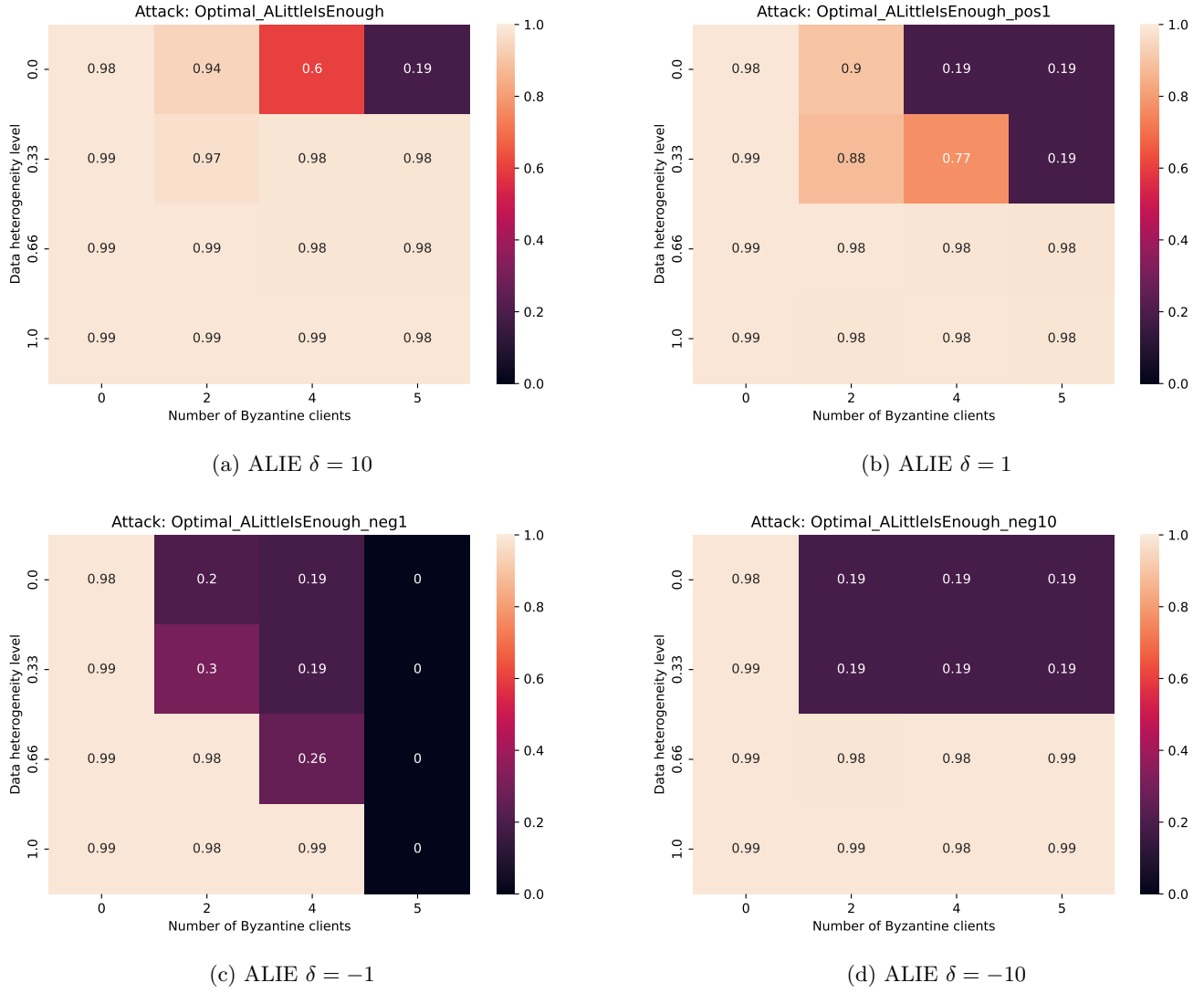


Figure 2: SNN accuracy comparison under ALIE attacks using the best overall aggregator selection.

3 ALIE Attack: Centered Clipping

This section compares performance under ALIE using Centered Clipping.

3.1 Convolutional Neural Network (CNN)

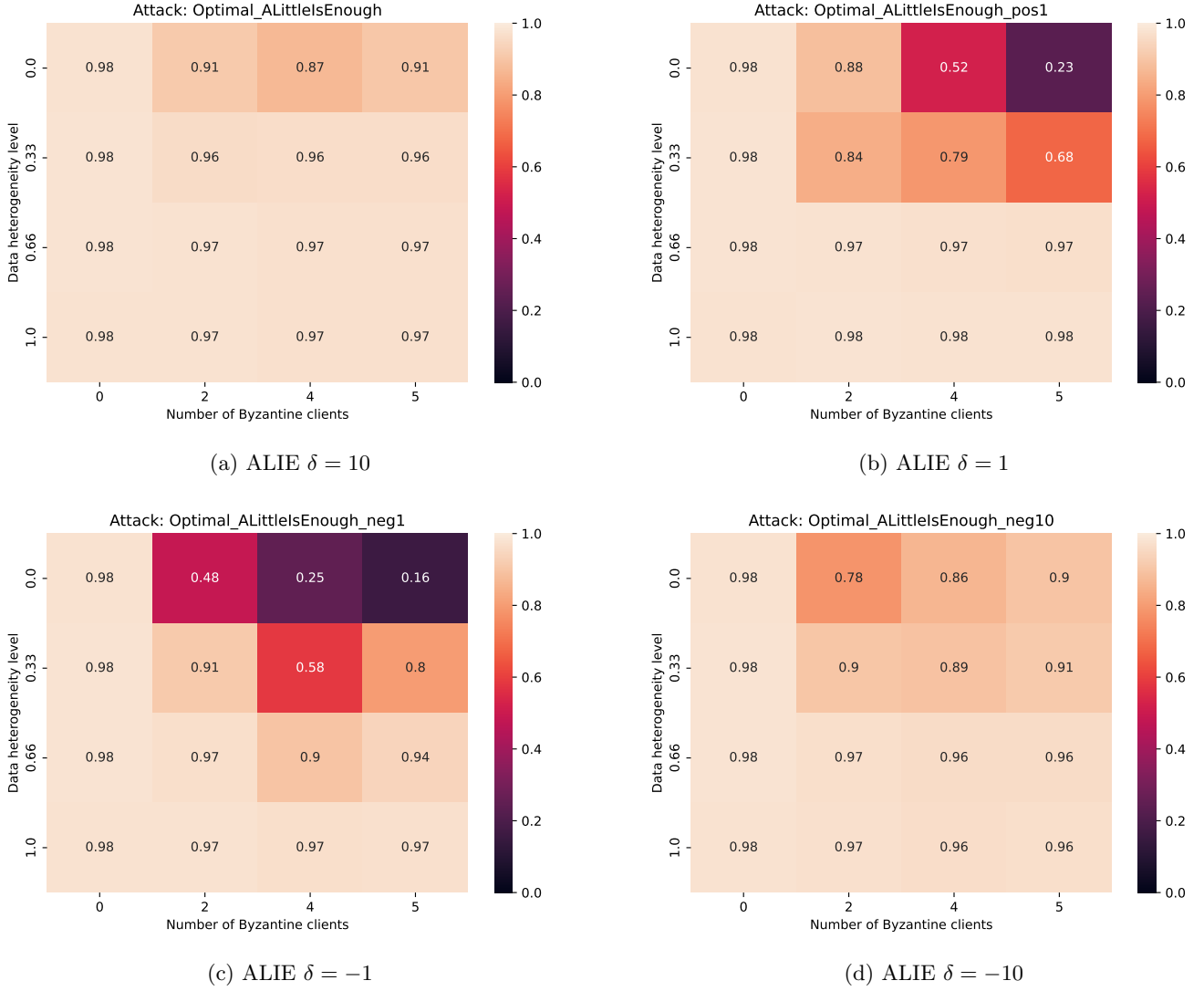
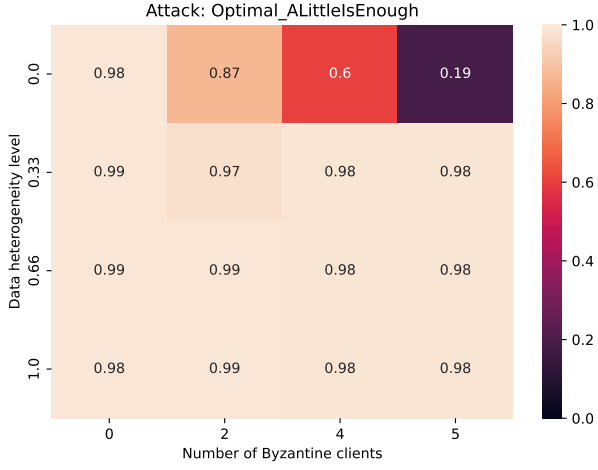
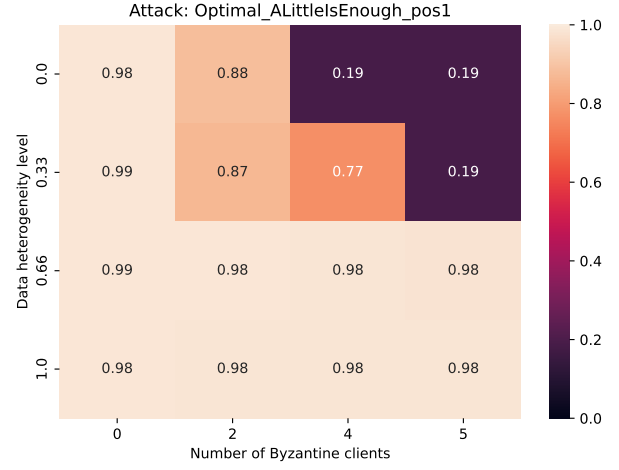


Figure 3: CNN accuracy comparison under ALIE attacks using Centered Clipping.

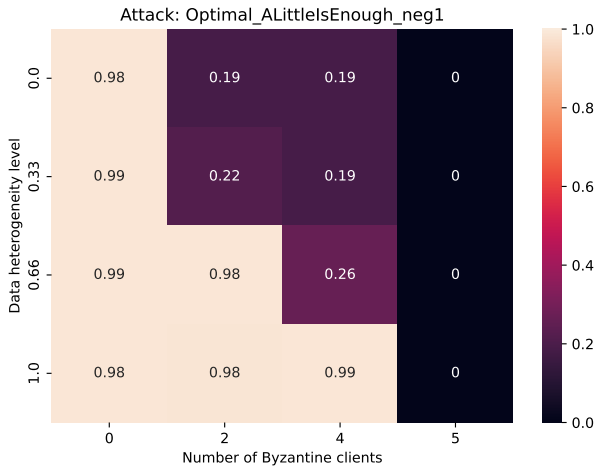
3.2 Spiking Neural Network (SNN)



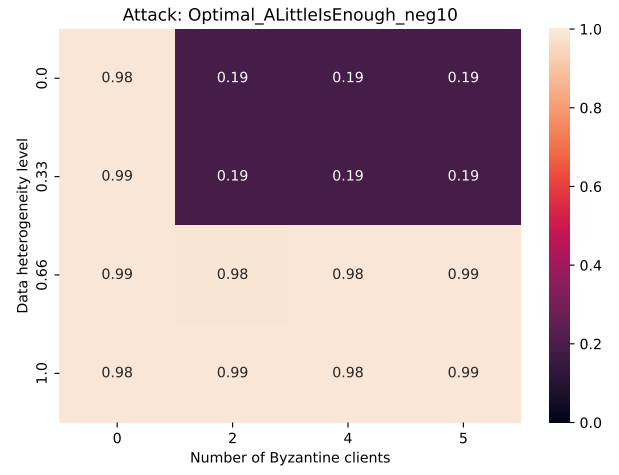
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$



(d) ALIE $\delta = -10$

Figure 4: SNN accuracy comparison under ALIE attacks using Centered Clipping.

4 ALIE Attack: Geometric Median

This section compares performance under ALIE using Geometric Median.

4.1 Convolutional Neural Network (CNN)

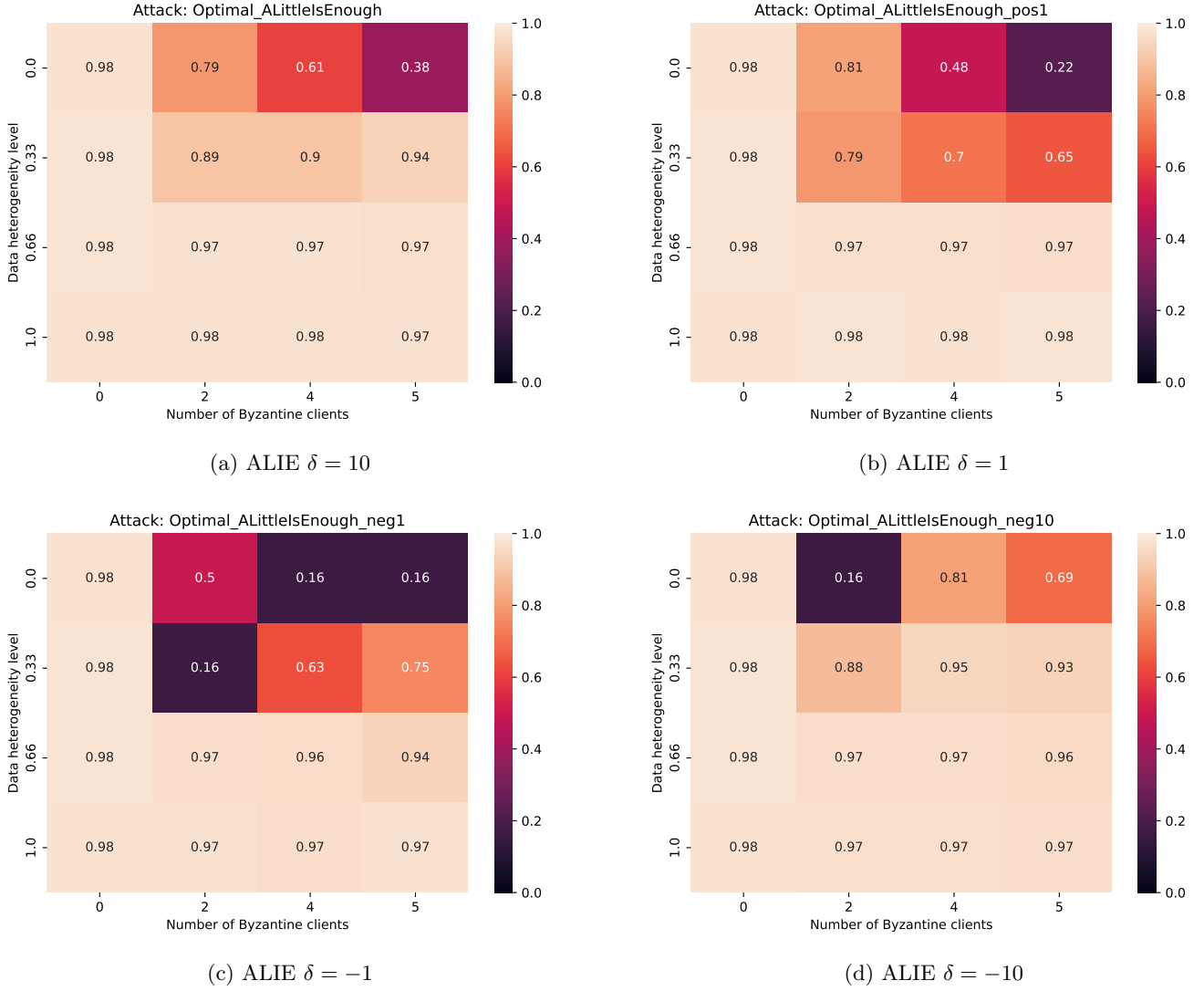
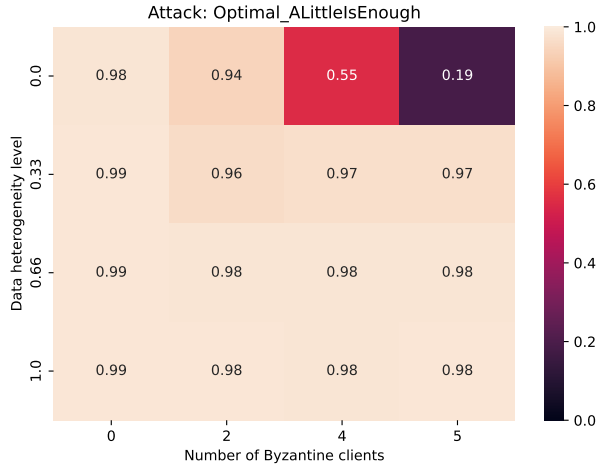
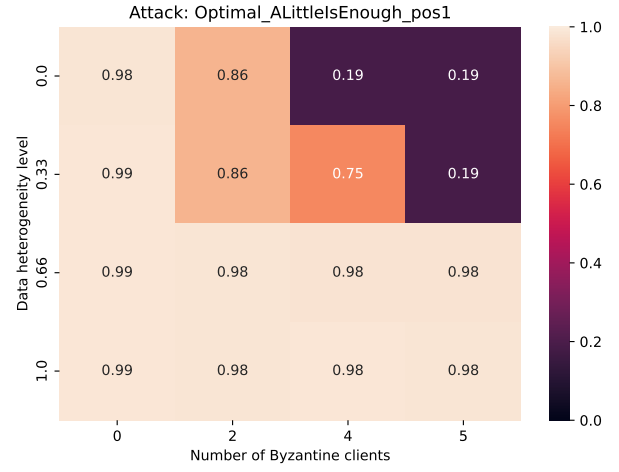


Figure 5: CNN accuracy comparison under ALIE attacks using Geometric Median.

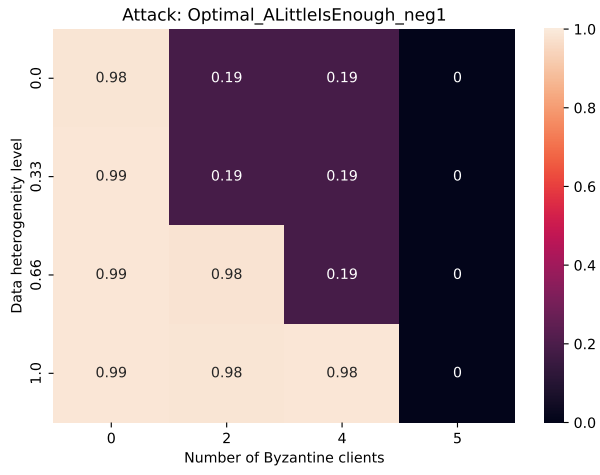
4.2 Spiking Neural Network (SNN)



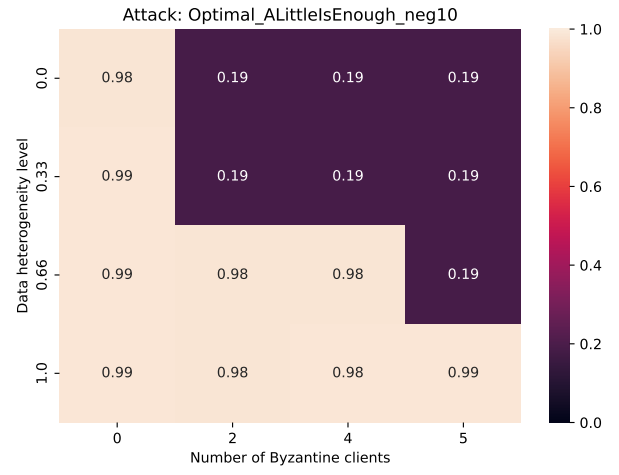
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$



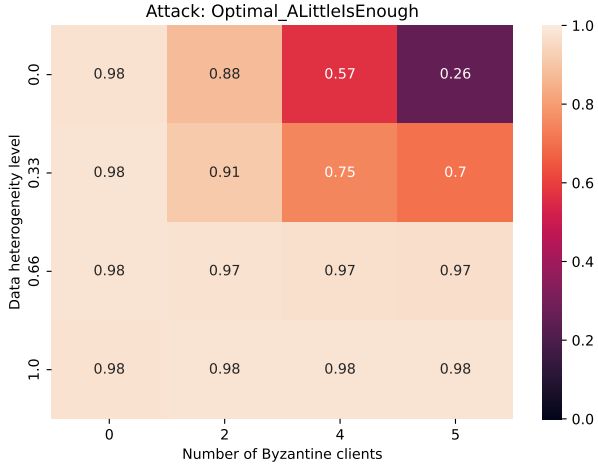
(d) ALIE $\delta = -10$

Figure 6: SNN accuracy comparison under ALIE attacks using Geometric Median.

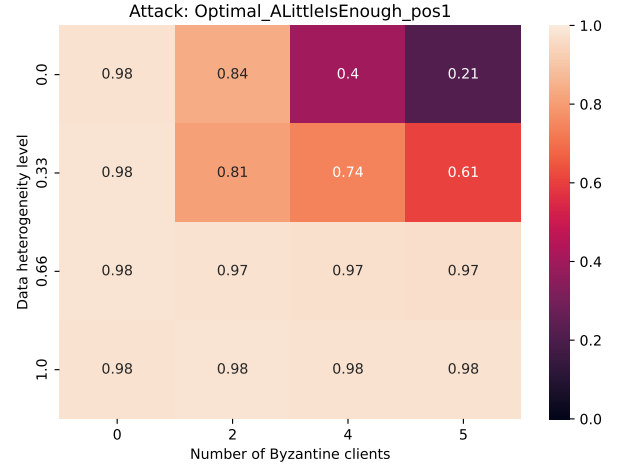
5 ALIE Attack: Multi-Krum

This section compares performance under ALIE using Multi-Krum.

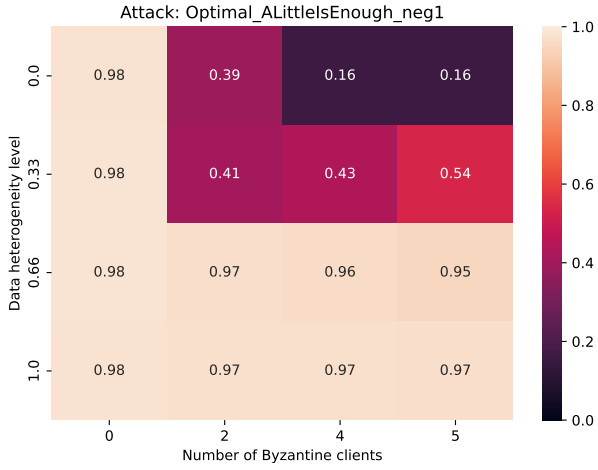
5.1 Convolutional Neural Network (CNN)



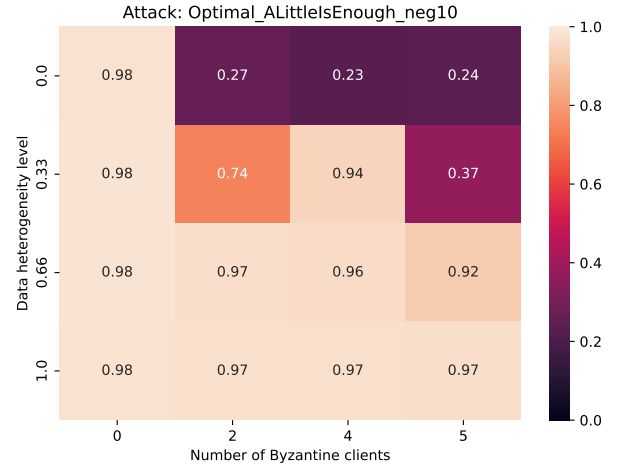
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



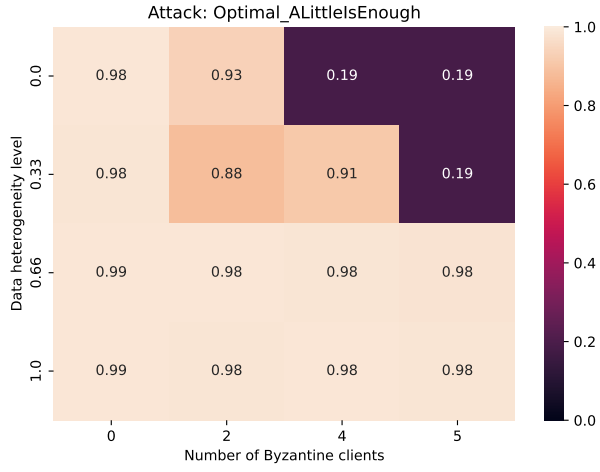
(c) ALIE $\delta = -1$



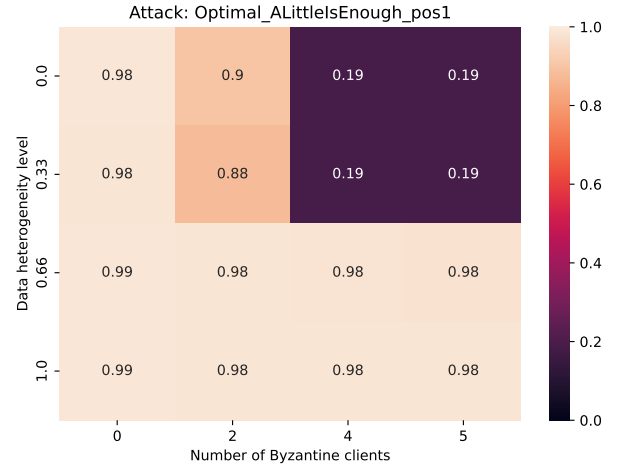
(d) ALIE $\delta = -10$

Figure 7: CNN accuracy comparison under ALIE attacks using Multi-Krum.

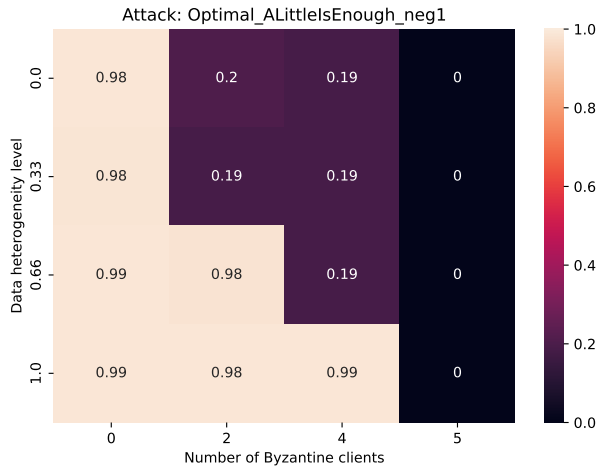
5.2 Spiking Neural Network (SNN)



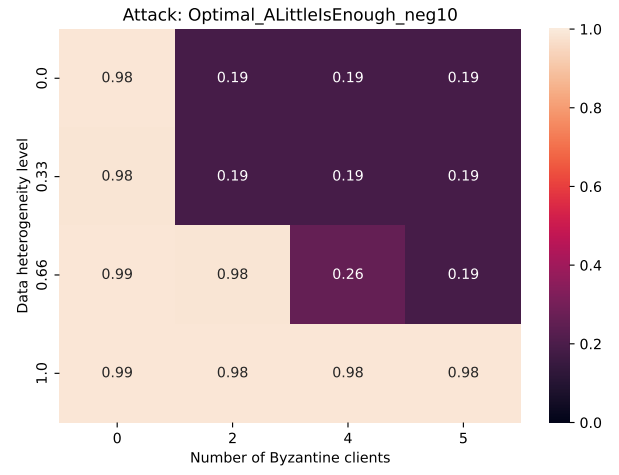
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$



(d) ALIE $\delta = -10$

Figure 8: SNN accuracy comparison under ALIE attacks using Multi-Krum.

6 ALIE Attack: Trimmed Mean

This section compares performance under ALIE using Trimmed Mean.

6.1 Convolutional Neural Network (CNN)

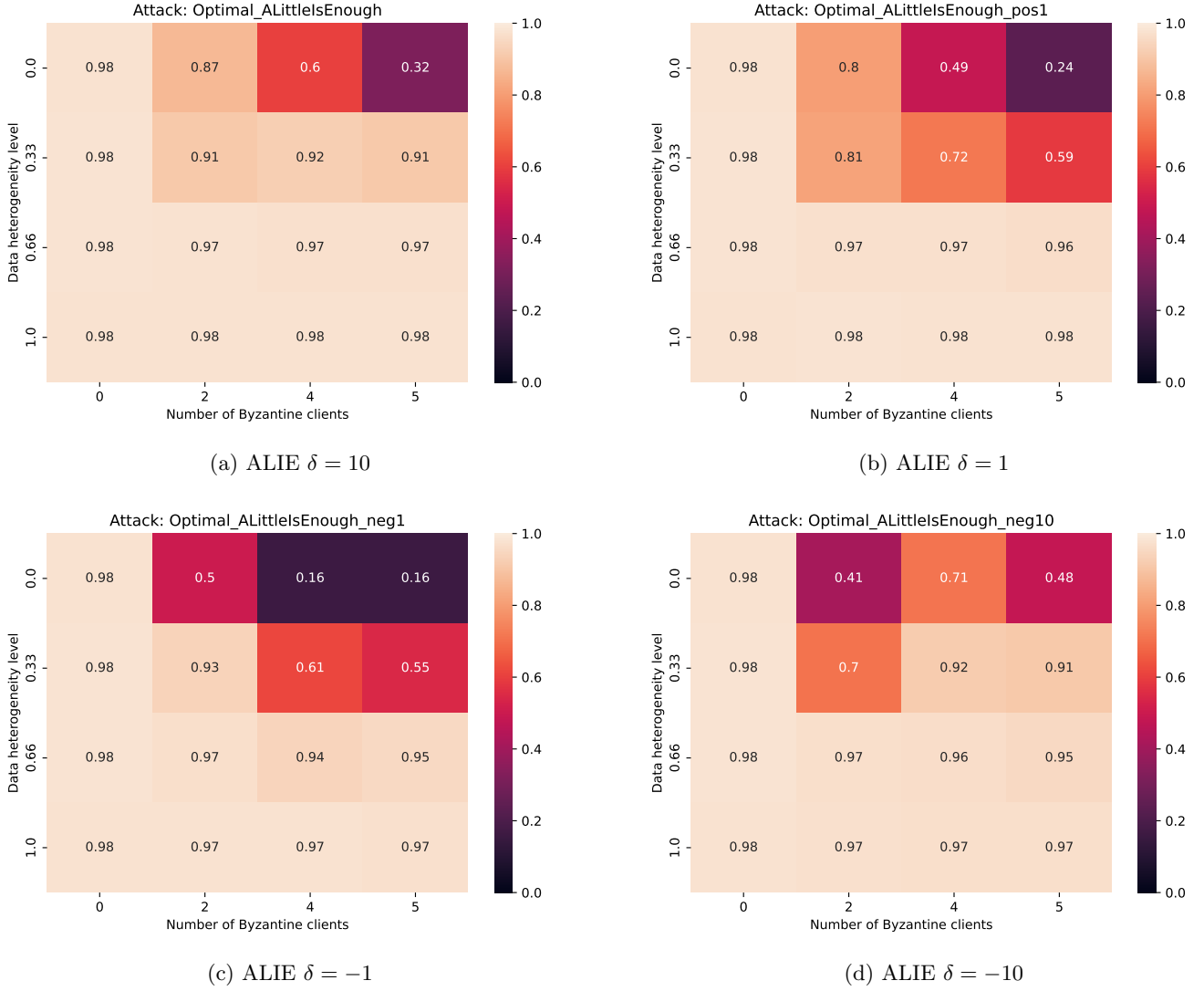
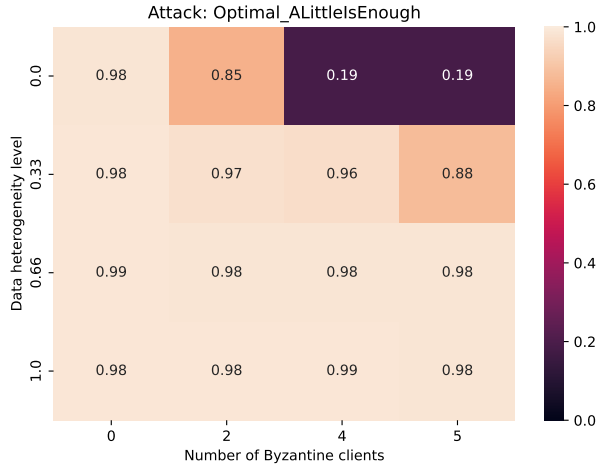
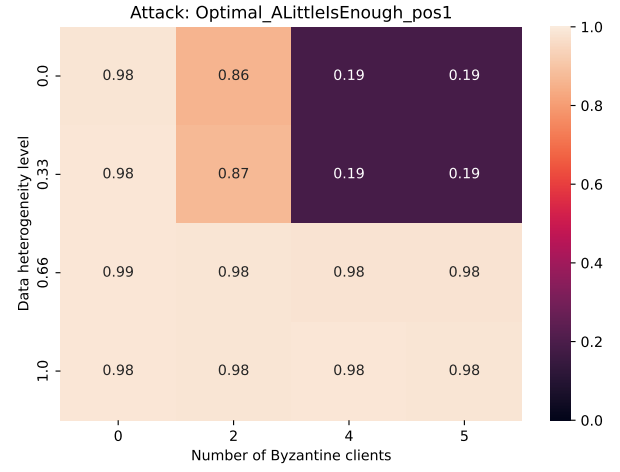


Figure 9: CNN accuracy comparison under ALIE attacks using Trimmed Mean.

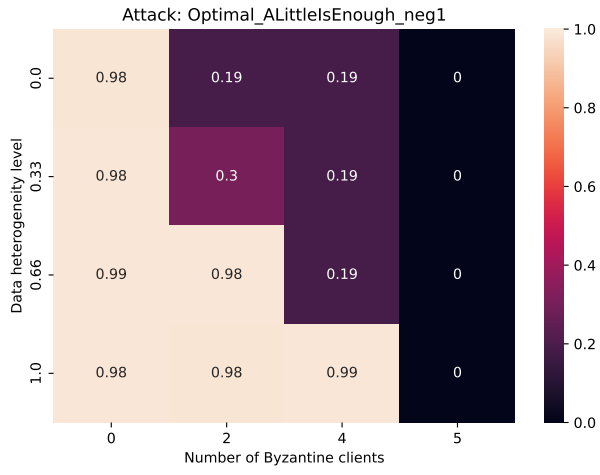
6.2 Spiking Neural Network (SNN)



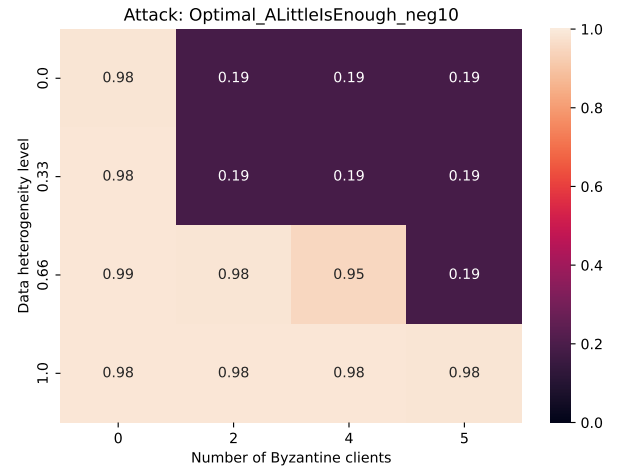
(a) ALIE $\delta = 10$



(b) ALIE $\delta = 1$



(c) ALIE $\delta = -1$



(d) ALIE $\delta = -10$

Figure 10: SNN accuracy comparison under ALIE attacks using Trimmed Mean.